

POOJITA SAI SUKANYA P

Email: pondaripss@gmail.com

LinkedIn: <https://www.linkedin.com/in/ps-p/>

Phone: 940-843-8595

GitHub: <https://github.com/PoojiPondari>

PROFFESIONAL SUMMARY

Innovative Full-Stack Developer with 3 years of experience designing and deploying scalable, AI-powered web applications. Led backend development using Python frameworks such as Django and Flask, automating complex workflows and optimizing data pipelines for analytics and predictive modeling. Engineered containerized microservices with Docker, managed orchestration via Kubernetes, and deployed resilient solutions on AWS and Azure, improving system scalability and uptime. Developed responsive, user-centric frontends using React.js and Angular, seamlessly integrating with RESTful APIs. Streamlined delivery cycles by implementing CI/CD pipelines with Jenkins, GitHub Actions, and GitLab CI, reducing deployment time and increasing code reliability. Applied AI/ML techniques for data-driven decision-making, leveraging tools like OpenAI APIs and scikit-learn for intelligent feature development. Consistently delivered clean, maintainable code in fast-paced Agile environments, driving product innovation and operational excellence.

WORKEXPERIENCE

AI-Engineer, Hartford

Jun 2024 - Present

- Led the end-to-end development and deployment of predictive machine learning models for insurance fraud detection, policyholder churn, and claim severity forecasting, using a combination of XGBoost, LightGBM, and TensorFlow, resulting in a 30% increase in detection accuracy and significant operational savings.
- Designed and productionized a regime-aware ML architecture using dual-model pipelines (bull/bear market classifiers), supporting real-time inference via AWS SageMaker, and enabling high-confidence financial signals for downstream automation.
- Architected and automated ML workflows using Apache Airflow, AWS Glue, and GitHub Actions, ensuring reliable data ingestion, feature engineering, model retraining, and CI/CD deployment across environments with full version control via DVC and experiment tracking via MLflow.
- Developed and deployed scalable, low-latency model APIs using Flask and Docker, integrated with AWS Lambda and SageMaker endpoints, supporting synchronous and asynchronous inference patterns for underwriting and risk operations.
- Built robust real-time monitoring and observability systems using Evidently AI, AWS CloudWatch, and Prometheus, detecting drift, anomaly spikes, and prediction inconsistencies, enabling proactive model governance and alerting for business-critical pipelines.
- Integrated secure access protocols using OAuth2, JWT, and fine-grained IAM policies, aligning ML systems with enterprise compliance (HIPAA, PII) and implementing audit-level logging for all inference and training activities.
- Led the creation of NLP-driven insights using Hugging Face Transformers, spaCy, and OpenAI GPT-4, extracting sentiment, intent, and high-risk signals from structured and unstructured claims data, increasing analytical coverage of complex cases by over 40%.
- Engineered a custom LLM-powered internal assistant using LangChain, FAISS, and OpenAI, enabling underwriters and analysts to query historical data, summarize policy documents, and generate scenario-driven risk narratives in natural language.
- Championed cross-functional collaboration with actuaries, product owners, DevOps, and compliance teams, delivering ML models in Agile sprints with comprehensive documentation (via GitHub and Confluence) and aligning model development with business objectives and regulatory standards.

Software Engineer -AI/ML, Infinity Connects Media

Jun 2022 – Jun 2023

- Designed and implemented a complete machine learning pipeline to detect fraudulent bookings, improving fraud detection accuracy by 28% and enabling proactive intervention.
- Collected and processed data from user activity logs, transaction records, and device metadata using Python and SQL; engineered 50+ features including time-based, behavioral, and aggregated metrics.
- Trained and optimized Logistic Regression, Random Forest, and XGBoost models using scikit-learn; addressed class imbalance with SMOTE and class-weight tuning to improve model recall.
- Deployed final models on Amazon SageMaker, enabling scalable, real-time inference with sub-150ms latency across the platform.
- Built Flask-based REST APIs for serving model predictions in both real-time and batch modes, supporting local testing and system integration.
- Automated the ML lifecycle with GitHub Actions, covering data validation, unit testing with PyTest, model evaluation, and SageMaker deployment in a repeatable CI/CD pipeline.
- Stored model versions, metadata, and logs in Amazon S3, ensuring reproducibility, rollback capability, and traceability of model performance.
- Developed custom monitoring scripts to detect model drift and input anomalies using statistical thresholds and SageMaker endpoint logs.
- Created a React.js dashboard for business users to visualize model outputs, risk scores, and confidence levels, enabling human-in-the-loop validation.
- Documented the entire ML workflow, from data processing to deployment and monitoring, in Confluence and GitHub Wiki, improving maintainability and team onboarding.

- Collaborated closely with product managers, DevOps, and data engineers in Agile sprints to deliver high-impact ML features in production and continuously refine model performance.

Software Developerp, Agile DataPro

Jun 2021– Jun 2022

- Led the development of scalable backend systems using Flask and Django, architecting secure RESTful APIs for user authentication, booking workflows, and payment gateways that handled thousands of concurrent users with high availability and low latency.
- Engineered full-stack web applications by integrating React.js and Redux with Python-based APIs, enabling real-time state updates, seamless client-server interaction, and consistent UX across desktop and mobile platforms.
- Optimized backend performance for high-throughput transactional systems by designing efficient MySQL data models, indexing strategies, and query optimizations—ensuring ACID compliance and millisecond-level response times under peak load.
- Applied machine learning models using scikit-learn (e.g., Logistic Regression, Random Forest) to enable intelligent user behavior analysis and predictive personalization features, improving customer engagement and retention rates.
- Built robust, automated CI/CD pipelines using Jenkins, Docker, and custom scripting to support continuous integration, testing, and deployment workflows across dev, staging, and production environments—cutting manual deployment efforts by 40%.
- Deployed containerized full-stack applications to AWS EC2, using Docker for environment consistency and custom shell scripts for provisioning, scaling, and monitoring cloud infrastructure in real-time.
- Established automated testing frameworks using pytest for backend logic and Postman collections for API contract validation, achieving high test coverage and reducing post-release regressions.

TECHNICAL SKILLS

Programming Languages: Python, JavaScript (Node.js), TypeScript, Bash, SQL

Frontend Frameworks: React.js, Angular, HTML5, CSS3, Redux, Tailwind CSS

Backend Frameworks: Django, Flask, Express.js (Node.js), Fast API, RESTful API Design

Cloud Platforms: AWS (EC2, S3, RDS, Lambda, SageMaker), Microsoft Azure, Google Cloud Platform (GCP)

DevOps & CI/CD: GitHub Actions, Jenkins, AWS Code Pipeline

Containerization & Orchestration: Docker, Kubernetes

Databases: MySQL, PostgreSQL, MongoDB

Version Control: Git, GitHub, Bitbucket, Confluence, Jira, Notion

Scripting & Automation: YAML, Shell Scripting

Security & Compliance: OAuth2, JWT, SSL/TLS, IAM (AWS Identity & Access Management)

Testing Frameworks: pytest, Postman, JUnit, Selenium

Machine Learning & AI: Scikit-learn, TensorFlow, XGBoost, LightGBM, Hugging Face, OpenAI API, LangChain\

Documentation & Collaboration: Confluence, Jira

CERTIFICATIONS

AWS: Certified Associate Developer, EKS Primer

Certified Kubernetes Administrator

Meta: Professional Backend Certificate

Microsoft Technology Associate (MTA): Programming using Python

Microsoft Technology Associate (MTA): Web Development using HTML, CSS, and Java

NPTEL: Introduction to the Internet of Things

Dev Town: Hotstar Clone using HTML, CSS, Bootstrap

Google: Introduction to Generative AI, Advanced Google Analytics

FreeCodeCamp: Machine Learning with Python

PROJECTS

PyPDFGen – Intelligent PDF Parsing & Prompt Automation Tool

Jan 2024 – May 2024

- Developed a Python-based PDF parsing tool utilizing regex, spaCy, and layout analysis to extract structured data from unstructured PDFs with over 90% accuracy, enhancing data accessibility for downstream processes.
- Integrated OpenAI's GPT-3.5 model to dynamically generate prompts based on the extracted content, automating summarization and streamlining decision-making workflows.
- Automated the entire process with modular scripts and scheduled jobs, significantly reducing manual analysis time by 70% and improving scalability across various PDF formats.

Predictive Model for Customer Churn Using Machine Learning

Aug 2023 – Dec 2023

- Conducted research under the guidance of a university professor to develop a predictive analytics solution for customer churn using Python (scikit-learn, Pandas), achieving over 85% accuracy with Logistic Regression and Random Forest models.
- Performed extensive exploratory data analysis (EDA) to identify key churn drivers, such as customer tenure and usage patterns, and built visual dashboards with Matplotlib and Seaborn to communicate insights.
- Documented the entire research process in an interactive Jupyter Notebook, delivering a modular, reproducible framework for future customer behavior prediction studies.

EDUCATION

Master's in Computer and Information Sciences

CGPA: 3.7

University of North Texas, Denton